TÉCNICO
UNIVERSIDADE
DE LISBOA

0.1 · COURSE PRESENTATION

Ship and Ocean Systems Design

Luís Batista

Instituto Superior Técnico / University of Lisbon

luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

COURSE OVERVIEW

Objectives

General

- Concept design
- Design process
- Laws and regulations
- Types of ships

Operational

- Main characteristics of the ship
- Optimization methods
- Hull form design
- Calculations and Specifications

Syllabus

1 **Design Process**
Design stages · Design spiral · Sequential and concurrent engineering · Actors · Design documents · Ship Product Data Model

2 **Ship Model – Initial Dimensioning**
Process of initial dimensioning · Estimation methods

3 **Voyage Model and Financial Model**
Voyage costs and time · Newbuilding cost · Ship financing · Operating costs

4 **Synthesis Model and Optimization in Ship Design**
Synthesis model · Single- and multi-objective optimization · MADM

5 **Design Criteria: Statutory Regulations and Classification Rules**
IMO Conventions: Load Lines, SOLAS, MARPOL, COLREG, Tonnage · ILO, OCIMF · EU Directives · National Laws

6 **Main Types of Merchant Ships**
Systems and equipment: bulk, container / multi-purpose, RoRo, tankers

7 **Hull Form, Lines Plan and General Arrangement**
Displacement distribution · Sections, bow, stern · Lines plan · GA

LEARNING OUTCOMES

After the Course, You Can…

1	**Frame a design problem**	Turn the owner's requirements (cargo, capacity, route, ports, speed, fuel, rules) into a design basis	**§1.3.5 · D1**
2	**Place the ship in the design process**	Name the design stages, actors and contract documents; classify the ship by its dominant constraint	**§1.3.1–1.3.7**
3	**Estimate the main characteristics**	L, B, D, T, CB, displacement, weights, power and capacities from regressions and methods	**Ch. 2, 5; App. A · D2–D3**
4	**Value the design**	Build a voyage model and a cost model: building cost, financing, operating cost	**§1.2; Ch. 6 · D4**
5	**Validate and check compliance**	Compare with reference ships; check freeboard, stability, EEDI and CII, draught at every port	**§2.16–2.19 · D5, D7**
6	**Optimize**	State variables, constraints and objectives; solve single- and multi-objective problems; choose with MADM	**§6.1; App. F · D6**
7	**Sketch the hull form and GA**	Turn the optimized dimensions into a hull form and a general arrangement, with a hydrostatics check	**Ch. 3–4 · D8**
8	**Recognize ship types**	Describe the systems and equipment of the main merchant ship types and the rules behind them	**Not tested**
9	**Specialize and defend**	Carry two specializations to a calculation note; present and justify the design in 15 minutes	**D9–D10**

Evaluation Method

Component	Description	Format	Weight
Evaluation Test	Design process; Law and Regulations reading	Multiple-choice	20%
Ship Design Project	Concept design of a merchant ship. Synthesis model and optimization.	Report	60%
Project Presentation	Presentation, discussion and justification of the Project.	Presentation	20%

Attendance and exercises: attendance is not compulsory but highly recommended. In-class exercises are individual practice on a reference ship, not graded; the project is group work done outside class.

Weekly self-assessment quizzes (ungraded). They show where you stand against the learning objectives and how deep the course goes.

Programming skills and software proficiency: 3D modelling, 2D drafting, programming and/or specific software use are neither taught nor evaluated in the present course. If you lack these skills, please consult the bibliography to reach a basic level of proficiency.

Academic integrity and use of AI. All submitted work must be your group's own. Copying from other groups, past reports or uncredited sources is plagiarism and will not be accepted. AI tools may be used as support, provided their use is declared in the report. You must be able to explain, justify and reproduce everything you submit; this is checked in the project presentation.

Lecture Calendar

Weekly timetable

Day	Time	Room
Wednesday	13:00–15:00	E2
Wednesday	15:00–16:30	E2
Friday	13:30–15:30	E2
Friday	15:30–17:00	E1

28 sessions over 7 weeks, from 4 November to 18 December.

Dates and milestones

Week	Date	Milestone
1	6 Nov	Project groups formed
4	27 Nov	Evaluation Test
6	13 Dec	Ship Design Project due
7	18 Dec	Project Presentation

The final version of the Ship Design Project will be formally delivered.

Class Structure

- "Types of Ship" runs as a continuous theme throughout the calendar.
- Each session ends with an individual exercise on a reference ship (not graded).

Office Hours

Sessions to discuss project

Book at calendly.com/vnsd-luisbatista/meetme

- 2 hours per week, in 30-minute slots, one group per slot.
- Each group may book no more than one slot per week.
- Upon scheduling, specify the purpose of the meeting and clearly state the points to be discussed.
- Location: Vera Navis Ship Design (VNSD), Av. 5 de Outubro nº 72 9º A

Course Plan · 28 Sessions

Week	Wed 13:00–15:00	Wed 15:00–16:30	Fri 13:30–15:30	Fri 15:30–17:00
Week 1 4–6 Nov	**W1.1** **Course + project**	**W1.2** Design process	**W1.3** Main dimensions	**W1.4** Design equation
Week 2 11–13 Nov	**W2.1** Human factors 1/2	**W2.2** Human factors 2/2	**W2.3** Dimensions · Rules	**W2.4** Power estimate
Week 3 18–20 Nov	**W3.1** Weights · deadweight	**W3.2** Hold capacity	**W3.3** Machinery · propeller	**W3.4** Voyage model
Week 4 25–27 Nov	**W4.1** Building cost	**W4.2** Financing · RFR	**W4.3** Stability · EEDI	**W4.4** **Evaluation Test**
Week 5 2–4 Dec	**W5.1** Synthesis model	**W5.2** Single-objective	**W5.3** Multi-objective	**W5.4** Hull form 1
Week 6 9–11 Dec	**W6.1** Hull form 2	**W6.2** Lines plan · GA	**W6.3** **Project clinic 1/2**	**W6.4** **Project clinic 2/2**
Week 7 16–18 Dec	**W7.1** Digital transition	**W7.2** AI in ship design	**W7.3** **Presentations 1/2**	**W7.4** **Presentations 2/2**

Design process · Ship model · Voyage and cost · Optimization and digital · Hull form and GA · Guest lecturer

Fixed: course, test, clinics, presentations

Week 1 · 4–6 November

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W1.1 **Wed 4 Nov** **13:00–15:00**	**Course presentation + Ship Design Project** Classification of merchant ships	0.1 and 0.2 [2]	**Exercise 1.1** Turn the reference ship's owner's brief into a design basis	**D1**
W1.2 **Wed 4 Nov** **15:00–16:30**	**Design process: phases, design spiral, owner's requirements** Deadweight, volume and linear-dimension carriers	§1.3.1–1.3.5 [1] · Shipbuilding contract [3]	**Exercise 1.2** Classify the ship and map its design spiral	**D1**
W1.3 **Fri 6 Nov** **13:30–15:30**	**Design methods; displacement estimate; main dimensions** Container carriers 1	§1.3.6–1.3.7; §2.1–2.2; App. A [1]	**Exercise 1.3** First Δ, L, B, T, D from the App. A regressions, with 95% bands Homework H0 (optional)	**D2**
W1.4 **Fri 6 Nov** **15:30–17:00**	**Displacement equation** Container carriers 2	§2.13; App. C (Normand) [1]	**Exercise 1.4** Displacement equation exercise, Q1–Q6 Homework H1 (optional)	**D3**

> **Milestone:** project groups formed, Fri 6 Nov.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] 0.1 Course Presentation and 0.2 Ship Design Project (course documents) · [3] Reading 1.2, Shipbuilding Contract (course reading).

Week 2 · 11–13 November

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W2.1 **Wed 11 Nov** **13:00–15:00**	**Human factors in ship design (1 of 2)** Guest lecturer	Guest lecturer's material	Set by the guest lecturer	—
W2.2 **Wed 11 Nov** **15:00–16:30**	**Human factors in ship design (2 of 2)** Guest lecturer	Guest lecturer's material	Set by the guest lecturer	—
W2.3 **Fri 13 Nov** **13:30–15:30**	**Length, beam, depth, draft, form coefficients · Law and Regulations introduction (20 min)** Containers, securing, hatch covers	§2.3–2.12; §2.7.1 [1]	**Exercise 2.3** Choose L, B, T, D, CB; check Fn, L/B, B/T, L/∇^(1/3) against the book's ranges	**D3**
W2.4 **Fri 13 Nov** **15:30–17:00**	**Preliminary power estimate** Multi-purpose vessels	§2.14; §5.1 [1]	**Exercise 2.4** Power by Admiralty and Holtrop–Mennen; compare	**D3**

> **Law and Regulations reading** starts on 13 Nov, for the Evaluation Test on 27 Nov.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

Week 3 · 18–20 November

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W3.1 **Wed 18 Nov** **13:00–15:00**	**Weights and centroids; deadweight; displacement check** Bulk carriers 1	§2.15–2.16 [1]	**Exercise 3.1** Lightship by two methods; close the weight balance	**D3**
W3.2 **Wed 18 Nov** **15:00–16:30**	**Hold capacity** Bulk carriers 2	§2.17 [1]	**Exercise 3.2** Hold volume against the target	**D3**
W3.3 **Fri 20 Nov** **13:30–15:30**	**Main machinery and propeller** RoRo ships	§5.2–5.3 [1]	**Exercise 3.3** Engine selection and propeller diameter Homework H2 (optional)	**D3**
W3.4 **Fri 20 Nov** **15:30–17:00**	**Voyage model** RoRo cargo and equipment	§1.2.1 [1] · Voyage notes A [2]	**Exercise 3.4** Round-trip time and fuel on the reference route	**D4**

> **Milestone:** ship model complete (D3).

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] Voyage and Finance Notes – Part A: The Voyage Model (course notes).

Week 4 · 25–27 November

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W4.1 **Wed 25 Nov** **13:00–15:00**	**Building cost** Oil tankers 1	§6.2–6.5 [1]	**Exercise 4.1** Newbuilding cost by weight group	**D4**
W4.2 **Wed 25 Nov** **15:00–16:30**	**Financing, operating costs, required freight rate** Oil tankers 2	Finance notes B [2] · How to finance a ship [3]	**Exercise 4.2** RFR and NPV; sensitivity to speed Homework H3 (optional)	**D4**
W4.3 **Fri 27 Nov** **13:30–15:30**	**Stability, trim, freeboard; EEDI** Product and chemical carriers	§2.18–2.19; §1.2.2 [1]	**Exercise 4.3** GM, freeboard and attained EEDI Homework H4 (optional)	**D7**
W4.4 **Fri 27 Nov** **15:30–17:00**	**Evaluation Test (20%)**	Law and Regulations reading [4]	Multiple choice	**Test**

Evaluation Test, Fri 27 Nov: design process and the Law and Regulations reading. Not tested: stability and freeboard, ship types, human factors.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.
[2] Voyage and Finance Notes, Part B · [3] Reading 4.2, How to Finance a Ship · [4] Law and Regulations handout

Week 5 · 2–4 December

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W5.1 **Wed 2 Dec** **13:00–15:00**	**The ship design optimization problem; ship synthesis model** General cargo ships and reefers	§6.1; App. F [1]	**Exercise 5.1** Link ship, voyage and cost models; tornado chart	**D6**
W5.2 **Wed 2 Dec** **15:00–16:30**	**Single-objective optimization** RoPax ferries	§6.1; App. F [1]	**Exercise 5.2** Optimization: lecturer demonstration, then an individual step Demo 5 · synthesis model	**D6**
W5.3 **Fri 4 Dec** **13:30–15:30**	**Multi-objective optimization and MADM** Fast ferries and catamarans	App. F (AFRAMAX case) [1]; TOPSIS [2]	**Exercise 5.3** Pareto front RFR vs EEDI; pick one design with TOPSIS	**D6**
W5.4 **Fri 4 Dec** **15:30–17:00**	**Hull form: distribution of displacement, waterlines** Passenger and cruise ships	§3.1–3.2 [1]	**Exercise 5.4** Sectional area curve and LCB Homework H5 (optional)	**D8**

Specialization checkpoint in office hours this week.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] Hwang, C. L. and Yoon, K. (1981). *Multiple Attribute Decision Making*. Springer.

Week 6 · 9–11 December

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W6.1 **Wed 9 Dec** **13:00–15:00**	**Hull form: sections, bow and bulbous bow, stern; hull series** Offshore tug and supply vessels	§3.3–3.5; App. B [1]	**Exercise 6.1** Sections, bow and stern; bulb or not (§3.4.2); parent hull from App. B	**D8**
W6.2 **Wed 9 Dec** **15:00–16:30**	**Lines plan, general arrangement, capacity plan** Fishing vessels	§4.1–4.6; §2.7.1 [1]	**Exercise 6.2** GA sketch: collision bulkhead, engine room, double bottom; capacity plan	**D8**
W6.3 **Fri 11 Dec** **13:30–15:30**	**Project clinic (1 of 2)**	Report and datasheet templates [2]	Review of each group's model and report	**Project**
W6.4 **Fri 11 Dec** **15:30–17:00**	**Project clinic (2 of 2)**	Report and datasheet templates [2]	Review of each group's model and report	**Project**

> **Ship Design Project due Sun 13 Dec (60%):** report and files, one ZIP by e-mail.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] 0.3 Report Template and 0.4 Datasheet Template (course documents).

Week 7 · 16–18 December

Session	Topic · ship types (10 min)	Reading	Individual exercise	Practice for
W7.1 **Wed 16 Dec** **13:00–15:00**	**Ship design in the digital transition** OBO carriers	App. G §G.2–G.6; App. F [1]	**Exercise 7.1** Sobol variants of L, B, D, CB through the synthesis model; RFR against EEDI (demonstration + individual step)	**D6**
W7.2 **Wed 16 Dec** **15:00–16:30**	**Artificial intelligence in ship design; AI demonstration** The groups' own ship types side by side	App. G §G.7–G.8 [1]	**Exercise 7.2** Surrogate model trained on Exercise 7.1's variants; re-optimize and check with the full model Demo 7 · surrogate model	**D6**
W7.3 **Fri 18 Dec** **13:30–15:30**	**Project presentations (1 of 2)**	—	15-min group presentation, then discussion	**Pres.**
W7.4 **Fri 18 Dec** **15:30–17:00**	**Project presentations (2 of 2)**	—	15-min group presentation, then discussion	**Pres.**

Project Presentation, Fri 18 Dec (20%): presentation, discussion and justification of the project, about 25 min per group.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

Bibliography

Alvariño, Ricardo; Azpíroz, Juan José y Meizoso, Manuel (1997). *El Proyecto Básico del Buque Mercante*. Fondo Editorial de Ingeniería Naval, Colegio de Ingenieros Navales.

Batista, Luis (2026). *Ship Design Lecture notes and Jupyter notebooks*.

Batista, Luis (2026). *3D Hull modelling manual and training videos in Rhinoceros*.

Buxton, Ian L. (1987). *Engineering Economics and Ship Design*, 3rd edition. British Maritime Technology.

Lamb, Thomas (2003). *Ship Design and Construction*. Vol. I, SNAME.

Lewis, E. V. (1988). *Principles of Naval Architecture*. Vols. I, II & III, SNAME.

Molland, Anthony F. (2008). *The Maritime Engineering Reference Book: A Guide to Ship Design, Construction and Operation*. Butterworth-Heinemann.

Navalapp. *Python programming notes*.

Papanikolaou, Apostolos (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd edition. Springer. Course textbook.

Roh, Myung-Il and Lee, Kyu-Yeul (2018). *Computational Ship Design*. Springer.

Schneekluth, H. and Bertram, V. (1998). *Ship Design for Efficiency and Economy*. Butterworth Heinemann.

Stopford, Martin (2009). *Maritime Economics*, 3rd edition. Routledge.

Ventura, Manuel (2021). *Lecture Notes of Ship and Ocean Systems Design*. Instituto Superior Técnico, University of Lisbon.